

At home water testing

From German brewing and more

Water composition is important for brewing and many brewers either send their water to a lab for analysis or build brewing water from scratch using very soft (e.g. reverse osmosis water) and salts. It is, however, also possible to test brewing water at home. The precision and amount of detail of such a water test does not match that of a professional analysis, but it is sufficient to estimate the residual alkalinity of the brewing water with an acceptable accuracy. At home water testing also allows regular testing of a water source in order to detect seasonal changes which may warrant a more precise professional analysis.

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Test kits

To test our water, we brewers can use water test kits available for aquarium owners. The kind of test kit we need is one that tests GH and KH. GH stands for General Hardness and measures the total hardness of the water. In water speak the total hardness is the amount of calcium and magnesium ions present. It is commonly measured as either dH (German Hardness) or "ppm as CaCO_3 ".

Both these units are equivalent measures. That means that they don't express the weight of the calcium and the magnesium ions but their number per unit of volume. With knowledge of their atomic weight and an guess on the calcium to magnesium ratio commonly found in water one can estimate the calcium and magnesium content of the analyzed water [DeLange].

KH stands for *Karbonathärte* (German for carbonate hardness) which is the alkalinity of the water. Like total hardness or GH it is measured as either German Hardness (dH) or "ppm as CaCO_3 ". The conversion of 1 dH = 17.8 CaCO_3 is true for both total hardness and alkalinity.

Two types test kits are available: titration based and strips. Titration based test kits contain test tubes and chemicals that are added until there is a color reaction in the water. Test strips are submersed in the water and the color reaction on their test pads is then compared against a color scale. While more difficult to use I prefer titration based tests since they provide more precise results are are easier to "read" than strips.



Figure 1 - A typical GH&KH test kit for aquarium use. It contains test tubes, test chemicals and instructions. The total price for kits like these is around \$6 and while there are more expensive test kits available for fish owners, they are useless to brewers since they test additional water parameters that aren't of interest for brewing.

Test procedure

For the exact instructions on how to use the test kit read the instruction that come with it. In particular how to convert the amount of titrant, that has been added, to a hardness value in dH or "ppm as CaCO_3 ". But in general they all work the same:

- rinse the test tube with the water to be tested.
- fill the test tube with a water sample up to its marking.
- add one drop of test solution, close the test tube and shake to mix water sample and test solution.
- repeat this process until the color suddenly changes.
- convert the number of drops that have been added to a hardness value using the instruction sheet.



Figure 2 - adding the titration solution to the water sample in the test tube one drop at a time

Titration

As mentioned earlier the process by which these tests work is commonly known as titration. During a titration a chemical, called titrant, is added to a sample of known volume. This titrant reacts with the compound to be tested until all of that compound in the sample has been consumed. An indicator shows when this point has been reached. Based on the amount of titrant that has been added the unknown amount of the compound in the tested solution can be calculated.

In the case of an alkalinity test (KH test) a strong acid is added which consumes the

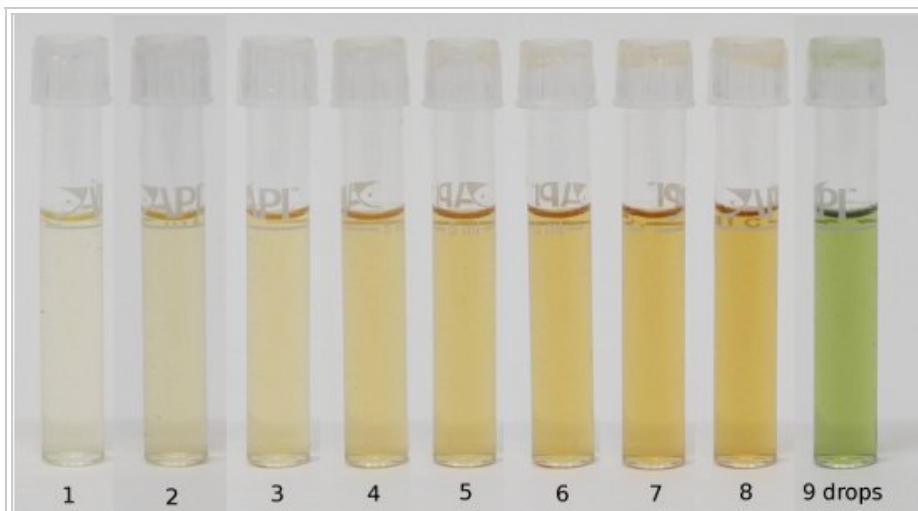


Figure 3 - GH titration of a water sample. It took 9 drops until the color changed from yellow to green. Thus the hardness of the water was about 9 dH or 160 ppm as CaCO_3

added which consumes the carbonates and bicarbonates. Those are the ions that establish the water's alkalinity. This reaction produces carbonic acids and more importantly lowers the pH. Once the pH has been lowered to about 4.3 virtually all carbonates and bicarbonates have been converted, at which point the indicator solution, which is nothing more than a pH indicator changes color from blue to yellow/green. The concentration of the test solution is designed such that each drop contains enough acid to neutralize 1 dH or 17.8 ppm as CaCO_3 in 5 ml water. As a result the precision of the test can be increased by increasing the sample size to 10 ml and assuming that each drop stands for 0.5 dH or 8.9 ppm as CaCO_3 .

A similar reaction takes place during testing for GH.

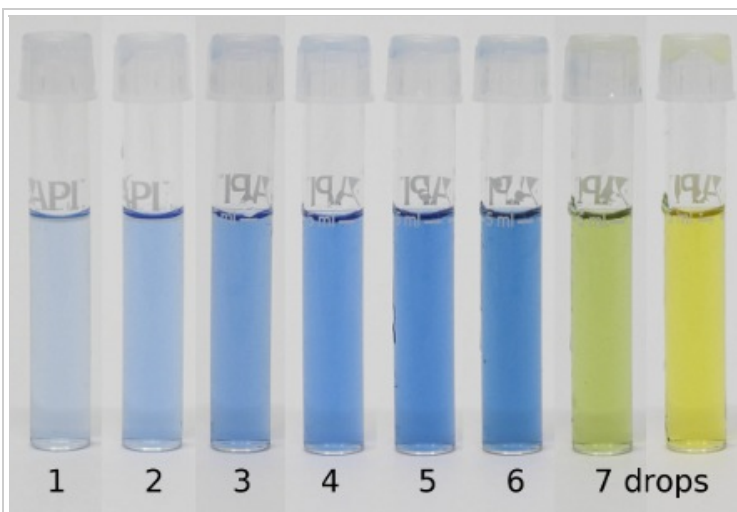


Figure 4 - KH titration of the same water sample. It took 7 drops until the color changed from blue to green and later yellow. Thus the alkalinity of the water is 7 dH or 125 ppm as CaCO_3

Estimating Residual Alkalinity

Just knowing alkalinity and total hardness of the water doesn't help for brewing. Brewers are interested in residual alkalinity which is determined by the water's alkalinity, its calcium content and to a lesser extent its magnesium content. Total hardness, however, lumps calcium and magnesium content together.

When looking at various water reports one will find that while the calcium to magnesium ratio is not constant on average about 30% of the total hardness is contributed by magnesium. This seems to be true for for most waters and Figure 5 illustrates that with data from various water profiles which brewers posted on-line (homebrewtalk.com).

Using that assumption residual alkalinity can be estimated from GH and KH as

$$\text{RA} = \text{KH} - \text{GH} / 4$$

This formula works with KH and GH given as either dH or "ppm as CaCO_3 ".

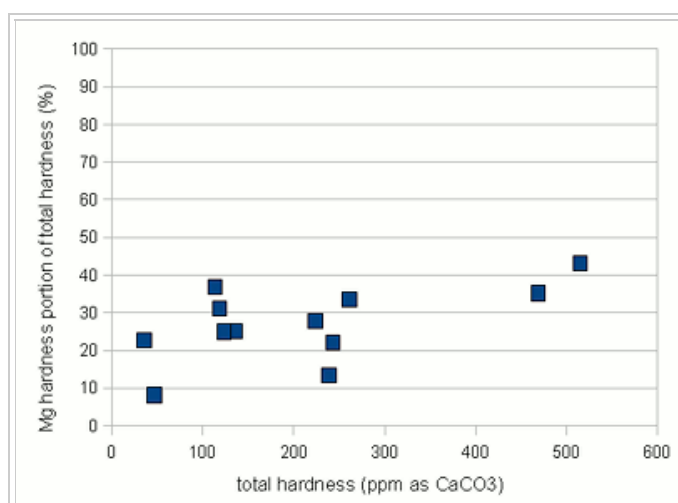


Figure 5 - Portion of magnesium hardness in percent plotted over the total hardness for a number of water profiles posted by members of the homebrewtalk.com (<http://www.homebrewtalk.com>) on-line forum

Using the test results in a spread sheet

If you don't want to calculate residual alkalinity by hand or want to estimate mash pH, you can use the water calculator (http://braukaiser.com/documents/Kaiser_water_calculator.xls). This is an Excel spread sheet that, among other features, allows entering KH and GH results from a simple water test. If you don't own Microsoft Excel check out OpenOffice (<http://www.openoffice.org>) which is a free office suite that supports excel spreadsheets. In fact, I develop all my spreadsheets with that software.



Step 1: enter the GH and KH test results.

simple water test			
	GH	KH	
	9	7	dH
or	160.2	124.6	ppm CaCO ₃
			calcium 45 ppm
			magnesium 12 ppm
			alkalinity 125 as CaCO ₃
			residual alkalinity 86 as CaCO ₃

If not entered as "ppm as CaCO₃" the measurement is converted to that unit. With the assumption that 30% of the total hardness are contributed by Magnesium the calcium and magnesium content of the water are calculated. No assumption needs to be made for calculating alkalinity. For the water used in the above example (GH = 9 dH; KH = 7 dH) the residual alkalinity is 86 ppm as CaCO₃.

base water and blending					
	Water A percentage		100	%	
					6.3 ion balance(%)
	Water A	Water B	analysis	mix	
calcium	44.98	0	mg/l	45.0	Ca mg/l
magnesium	11.69	0	mg/l	11.7	Mg mg/l
sodium		0	mg/l	0.0	Na mg/l
sulfate		0	mg/l	0.0	SO ₄ mg/l
chloride		0	mg/l	0.0	Cl mg/l
bicarb		0	mg/l	152.0	HCO ₃ mg/l
alkalinity	124.6	0	ppm CaCO ₃	124.6	CaCO ₃ mg/l
			Residual alkalinity	86	as CaCO ₃

If no other values are given in the field for a more detailed water report, the results from the GH & KH test are carried over and will be used by the rest of the spreadsheet. Note that no data is available for Sulfate or Chloride content.

Water use		
total water	30	l
Strike water	12	l
Sparge water	18	l

malt weight		
grist weight	4	kg
mash thickness	3	l/kg

beer color based mash pH estimation					
beer color	15	SRM			target pH
roasted %	50	%		5.2	5.4
estimated mash pH	5.5	RA (as ppm CaCO ₃)	-160	33	225

Step 2: After strike water volume, grist weight and beer color have been specified an mash pH estimate is made based on these parameters and the residual alkalinity of the water. "Roasted %" refers specifies the percentage of specialty malts which are roasted malts (e.g. Roasted Barley, chocolate, Carafo, black patent and others). In this example the beer color was 15 SRM and half the specialty malts were roasted malts. The mash pH is likely to be around 5.5 which is a proper pH for mashing and no water treatment is necessary.

If water treatment is necessary these techniques are supported by the spreadsheet: blending, salt and acid addition.

Acknowledement

My acknowledgement goes to A.J. deLange who pointed me to the idea of home water testing in his article Understanding Alkalinity and Hardness (http://braukaiser.com/documents/not_mine/deLange_Alkalinity_I.pdf).

References

[deLange] A.J. deLange, "Understanding Alkalinity and Hardness Part I", ajdel.wetnewf.org (<http://ajdel.wetnewf.org:81/>)

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